

ECEN 635 Note Set #2

General Solution for uniform plane wave

uniform $\Rightarrow$ equi-phase planes are also equi-amplitude planes

The complete free space solution to $\nabla^2 \mathbf{E} + k^2 \mathbf{E} = 0$ is

$$\mathbf{E}(\mathbf{r}) = \mathbf{E}_0 e^{-j\mathbf{k} \cdot \mathbf{r}} = \mathbf{E}_0 e^{-jk\hat{\mathbf{n}} \cdot \mathbf{r}}$$

where

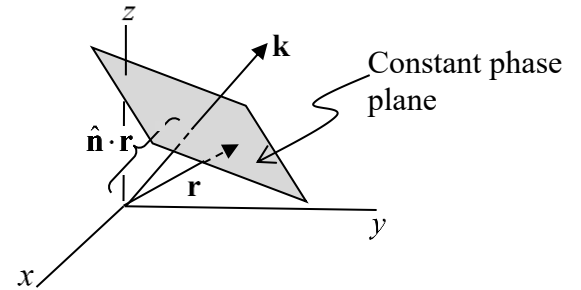
$$\mathbf{r} = x\hat{\mathbf{x}} + y\hat{\mathbf{y}} + z\hat{\mathbf{z}}$$

is a position vector of the observation point (x, y, z)

$$\begin{aligned} \mathbf{k} &= k_x \hat{\mathbf{x}} + k_y \hat{\mathbf{y}} + k_z \hat{\mathbf{z}} \\ &= k \hat{\mathbf{n}} \end{aligned}$$

$\mathbf{k}$ is a vector in the direction of propagation of the plane wave, called the propagation vector or wave number.

$\hat{\mathbf{n}}$ is a unit vector in the direction of propagation.



Notice: $k = \sqrt{k_x^2 + k_y^2 + k_z^2}$

- Proof that the electric field, magnetic field and $\mathbf{k}$ are $\perp$ to each other.

In a source free region

$$\begin{aligned} \nabla \cdot \mathbf{E} &= 0 \\ \nabla \cdot (\mathbf{E}_0 e^{-j\mathbf{k} \cdot \mathbf{r}}) &= e^{-j\mathbf{k} \cdot \mathbf{r}} \nabla \cdot \mathbf{E}_0 + \mathbf{E}_0 \cdot (\nabla e^{-j\mathbf{k} \cdot \mathbf{r}}) = 0 \\ \mathbf{E}_0 \cdot \left(\nabla e^{-j(k_x x + k_y y + k_z z)} \right) &= 0 \\ \mathbf{E}_0 \cdot (-jk_x \hat{\mathbf{x}} - jk_y \hat{\mathbf{y}} - jk_z \hat{\mathbf{z}}) e^{-j\mathbf{k} \cdot \mathbf{r}} &= 0 \\ \mathbf{E}_0 \cdot (-j\mathbf{k} e^{-j\mathbf{k} \cdot \mathbf{r}}) &= 0 \\ \mathbf{E}_0 \cdot \mathbf{k} &= 0 \\ \text{or} \\ \mathbf{E}_0 \cdot \hat{\mathbf{n}} &= 0 \end{aligned}$$

$\therefore$ In a source free space the electric field is $\perp$ the direction of propagation.

Also in the same way from $\nabla \cdot \mathbf{H} = 0$ we get

$$\mathbf{H}_0 \cdot \hat{\mathbf{n}} = 0$$

So in a source free space the magnetic field is $\perp$ the direction of propagation. Waves where $\mathbf{E}$ and $\mathbf{H}$ are $\perp$ the direction of propagation are called TEM (transverse electromagnetic) waves.

‘Short form’ for Maxwell’s Equations – For plane waves and all other TEM (transverse electromagnetic) waves only.

$$\begin{aligned}
 \nabla \times \mathbf{E} &= -j\omega\mu\mathbf{H} \\
 \Rightarrow \mathbf{H} &= \frac{j}{\omega\mu} \nabla \times \mathbf{E} \\
 &= \frac{j}{\omega\mu} \nabla \times \mathbf{E}_0 e^{-j\mathbf{k} \cdot \mathbf{r}} \\
 &= \frac{j}{\omega\mu} \left\{ e^{-j\mathbf{k} \cdot \mathbf{r}} \nabla \times \mathbf{E}_0 - \mathbf{E}_0 \times \nabla e^{-j\mathbf{k} \cdot \mathbf{r}} \right\} \\
 &= \frac{j}{\omega\mu} (-\mathbf{E}_0) \times \underbrace{(-j\mathbf{k} e^{-j\mathbf{k} \cdot \mathbf{r}})}_{\text{from previous page}} \\
 &= \frac{+1}{\omega\mu} (\mathbf{k} \times \mathbf{E}_0 e^{-j\mathbf{k} \cdot \mathbf{r}}) \\
 \mathbf{H} &= \frac{1}{\omega\mu} \mathbf{k} \times \mathbf{E} = \frac{k}{\omega\mu} \hat{\mathbf{n}} \times \mathbf{E}
 \end{aligned}$$

Since in the previous calculation it was shown that $\mathbf{E}$ and $\mathbf{H}$ are $\perp$ to direction of propagation $\hat{\mathbf{n}}$, the conclusion with the above calculation is that $\mathbf{E}$ and $\mathbf{H}$ are $\perp$ to each other.

It was also shown above that if the plane wave $\mathbf{E}$ is known then $\mathbf{H}$ can be found from

$$\mathbf{H} = \frac{j}{\omega\mu} \nabla \times \mathbf{E} = \frac{k}{\omega\mu} \hat{\mathbf{n}} \times \mathbf{E}$$

Also it can be shown from

$$\nabla \times \mathbf{H} = -j\omega\epsilon\mathbf{E} \quad \text{that} \quad \mathbf{E} = \frac{k}{\omega\epsilon} \mathbf{H} \times \hat{\mathbf{n}}$$

$\therefore$ In a homogeneous lossless medium

$$\begin{aligned}
 \mathbf{H} &= \frac{k}{\omega\mu} \hat{\mathbf{n}} \times \mathbf{E} = \frac{1}{\eta} \hat{\mathbf{n}} \times \mathbf{E} \\
 \mathbf{E} &= \frac{k}{\omega\epsilon} \mathbf{H} \times \hat{\mathbf{n}} = -\eta \hat{\mathbf{n}} \times \mathbf{H}
 \end{aligned}$$

where $\eta = \sqrt{\frac{\mu}{\epsilon}}$

Wave polarization

Assume a plane wave traveling in the $+z$ direction

$$\begin{aligned}\mathbf{E} &= (E_{x0}\hat{\mathbf{x}} + E_{y0}\hat{\mathbf{y}})e^{-jkz} \\ &= (|E_{x0}|e^{j\theta_x}\hat{\mathbf{x}} + |E_{y0}|e^{j\theta_y}\hat{\mathbf{y}})e^{-jkz}\end{aligned}$$

In the time domain and suppressing the $\sqrt{2}$ out front

$$\mathcal{E}(t) = |E_{x0}|\cos(\omega t - kz + \theta_x)\hat{\mathbf{x}} + |E_{y0}|\cos(\omega t - kz + \theta_y)\hat{\mathbf{y}}$$

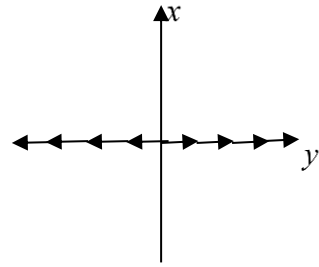
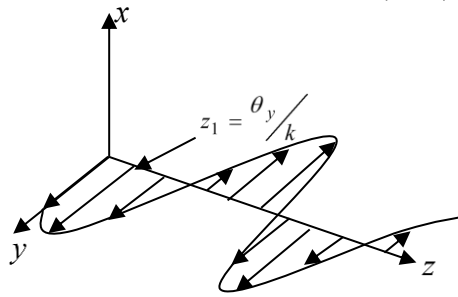
Case 1 Linear Polarization

(i) $\theta_x = \theta_y$

(ii) $|E_{x0}| = A|E_{y0}|$

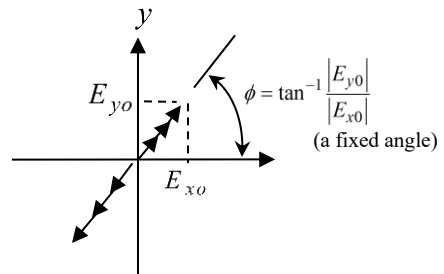
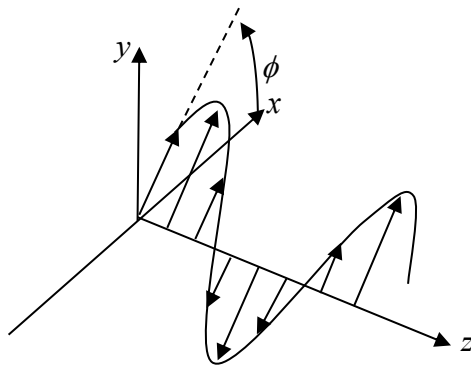
Example 1: Let $|E_{x0}| = 0$

$$\Rightarrow \mathcal{E}(t) = |E_{y0}|\cos(\omega t - kz + \theta_y)\hat{\mathbf{y}}$$



Example 2:

Let $E_{x0} \neq E_{y0}$, $\theta_x = \theta_y$



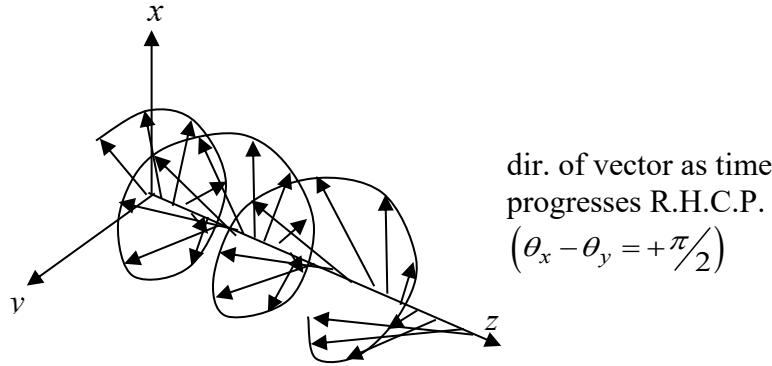
Case 2 Circular Polarization

$$|E_{x0}| = |E_{y0}| = E_0, \theta_x = \theta_y \pm \pi/2$$

Proof:

$$\begin{aligned} \mathcal{E}^2(t) &= \mathcal{E} \cdot \mathcal{E} = \mathcal{E}_x^2 + \mathcal{E}_y^2 \\ &= E_0^2 \cos^2(\omega t - kz + \theta_x) + E_0^2 \cos^2(\omega t - kz + \theta_x \pm \pi/2) \\ &= E_0^2 \cos^2(\omega t - kz + \theta_x) + E_0^2 \sin^2(\omega t - kz + \theta_x) \\ &= E_0^2 \end{aligned}$$

$\mathcal{E}_x^2 + \mathcal{E}_y^2 = E_0^2$ is the equation of a circle



Case 3 Elliptical Polarization

(a) $|E_{x0}| \neq |E_{y0}|, \theta_x = \theta_y \pm \pi/2$

(b) $|E_{x0}| \neq |E_{y0}|, \theta_x \neq \theta_y$ and $\theta_x \neq \theta_y \pm \pi/2$

In the time domain $\mathcal{E}(t) = |E_{x0}| \cos(\omega t - kz + \theta_x) \hat{x} + |E_{y0}| \cos(\omega t - kz + \theta_y) \hat{y}$ viewed with the z-axis pointing directly at you:

Case 1 Linear Polarization

Circular Polarization

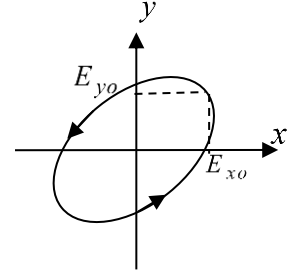
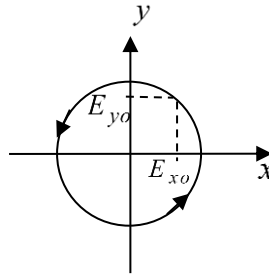
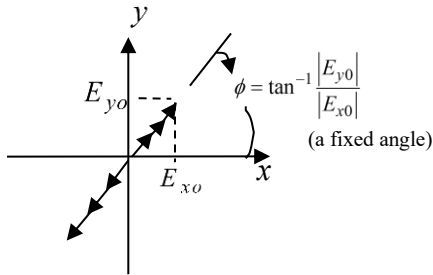
Elliptical Polarization

:

Let $E_{x0} \neq E_{y0}, \theta_x = \theta_y$

$|E_{x0}| = |E_{y0}| = E_0, \theta_x = \theta_y \pm \pi/2$

$|E_{x0}| \neq |E_{y0}|, \theta_x \neq \theta_y$ and $\theta_x \neq \theta_y \pm \pi/2$



Boundary Conditions

Scalar form

$$E_{t1} - E_{t2} = 0$$

$$H_{t1} - H_{t2} = J_s$$

$$\varepsilon_1 E_{n1} - \varepsilon_2 E_{n2} = \rho_s$$

$$\mu_1 H_{n1} - \mu_2 H_{n2} = 0$$

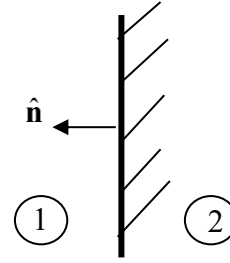
General vector form

$$\hat{\mathbf{n}} \times (\mathbf{E}_1 - \mathbf{E}_2) = 0$$

$$\hat{\mathbf{n}} \times (\mathbf{H}_1 - \mathbf{H}_2) = \mathbf{J}_s$$

$$\hat{\mathbf{n}} \cdot (\varepsilon_1 \mathbf{E}_1 - \varepsilon_2 \mathbf{E}_2) = \rho_s$$

$$\hat{\mathbf{n}} \cdot (\mu_1 \mathbf{H}_1 - \mu_2 \mathbf{H}_2) = 0$$



For a perfect conductor skin depth

$$\delta = \sqrt{\frac{2}{\omega \mu \sigma}} \rightarrow 0$$

$\Rightarrow$ no fields in the perfect conductor (region 2)

$$\mathbf{E}_{t1} = \hat{\mathbf{n}} \times \mathbf{E}_1 = 0$$

$$\mathbf{H}_{t1} = \hat{\mathbf{n}} \times \mathbf{H}_1 = \mathbf{J}_s$$

$$\varepsilon_1 E_{n1} = \varepsilon_1 \hat{\mathbf{n}} \cdot \mathbf{E}_1 = \rho_s$$

$$\mu_1 H_{n1} = \mu_1 \hat{\mathbf{n}} \cdot \mathbf{H}_1 = 0$$

Deriving Boundary Conditions*

Tangential fields

From Maxwell's equations we have

$$\oint \mathbf{H} \cdot d\mathbf{l} = j\omega \int \mathbf{D} \cdot d\mathbf{s} + I \quad (1)$$

or

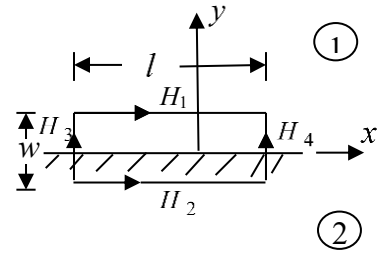
$$\nabla \times \mathbf{H} = j\omega \mathbf{D} + \mathbf{J} \quad (2)$$

Usually the boundary conditions are found by applying (1) on a closed line path that passes across the boundary.

However a second method comes from discretizing (2) as follows: If w and l are small (2) is approximated by

$$\frac{H_4 - H_3}{l} - \frac{H_1 - H_2}{w} = J_z + j\omega D_z$$

$$H_2 - H_1 = J_z w + \left(j\omega D_z - \frac{H_4 - H_3}{l} \right) w$$



Let l remain constant and take the limit $w \rightarrow 0$

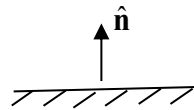
$$\lim_{w \rightarrow 0} J_z w = J_s \quad (\text{surface current } A/m)$$

$$\lim_{w \rightarrow 0} \left(j\omega D_z - \frac{H_4 - H_3}{l} \right) w = 0$$

$$\Rightarrow H_2 - H_1 = J_s$$

in general this equation can be written

$$\hat{\mathbf{n}} \times (\mathbf{H}_1 - \mathbf{H}_2) = \mathbf{J}_s$$



$\hat{\mathbf{n}}$ is normal to the surface pointing into region 1

The methods described above can be used on Maxwell's equation $\nabla \times \mathbf{E} = -j\omega \mathbf{B} - \mathbf{M}$, to get

$$\hat{\mathbf{n}} \times (\mathbf{E}_1 - \mathbf{E}_2) = -\mathbf{M}_s$$

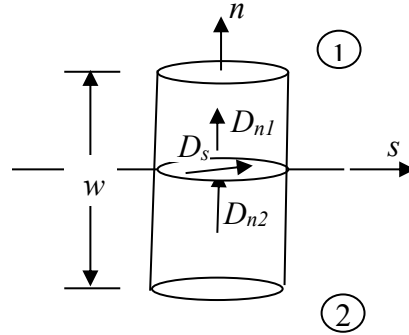
*This 'derivation' is not completely sound mathematically – see Stratton for a more complete method.

Normal fields

$$\nabla \cdot \mathbf{D} = \rho$$

$$\frac{\partial D_n}{\partial n} + \frac{\partial D_s}{\partial s} = \rho$$

$$\frac{D_{n1} - D_{n2}}{w} = \rho - \frac{\partial D_s}{\partial s}$$



multiply by w and take the limit as $w \rightarrow 0$

$$\lim_{w \rightarrow 0} \left[D_{n1} - D_{n2} = \rho w - \frac{\partial D_s}{\partial s} w \right]$$

On the right hand side

$$\lim_{w \rightarrow 0} \rho w = \rho_s \quad (\text{surface charge } C/m^2)$$

and

$$\lim_{w \rightarrow 0} \frac{\partial D}{\partial s} w = 0$$

resulting in

$$D_{n1} - D_{n2} = \rho_s$$

or more generally

$$\hat{\mathbf{n}} \cdot (\mathbf{D}_1 - \mathbf{D}_2) = \rho_s$$

The methods described above can be used on Maxwell's equation, $\nabla \cdot \mathbf{B} = \rho_m$ to get

$$\hat{\mathbf{n}} \cdot (\mathbf{B}_1 - \mathbf{B}_2) = \rho_{m,s}$$